EE599 Research Project
3. Device Design and Implementation

Hari Krishna Koneti
M.S. Electrical Engineering

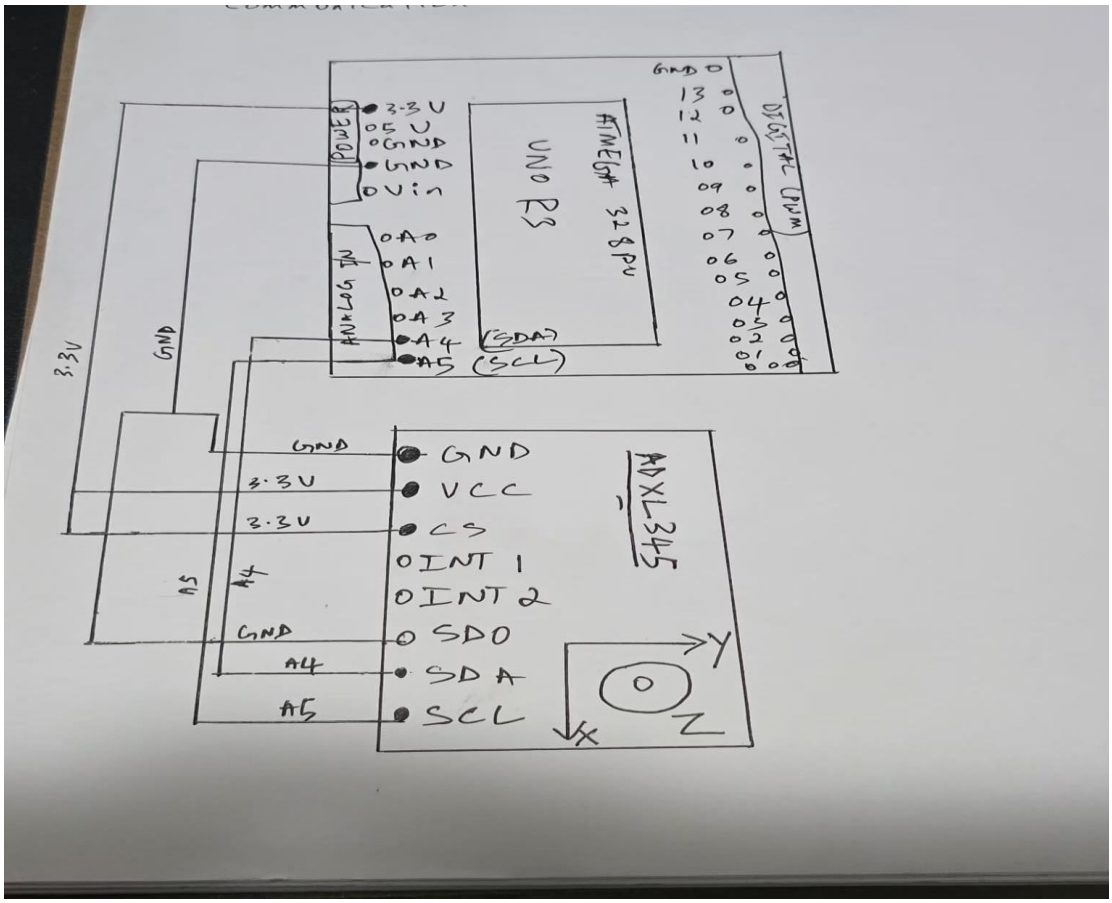
3. Device Design and Implementation

PostureSense is a wearable concept based on Arduino Uno R3 and Spark Fun ADXL345 to distinguish between sleep positions (supine, prone, left, right). The design is simple, inexpensive, and dependable..

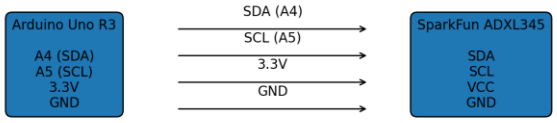
3.1 Hardware Design

Arduino Uno R3 performs acquisition and classification. ADXL345 measures 3-axis acceleration. System is powered via USB from laptop for development.

3.2 Electrical Connections and Schematic



I2C connections: SDA→A4, SCL→A5, VCC→3.3V, GND→GND.



3.2.1 Pin Description and Circuit Explanation

The Arduino Uno R3 and SparkFun ADXL345 are connected using the I2C communication interface. Each pin in the circuit serves a specific role in ensuring proper data transmission and power delivery.

Pin-Level Description

ADXL345 Pins

- VCC (3.3V)
  Sources the ADXL345 sensor. The sensor is powered by 3.3V and feeding it with more power (e.g., 5V), will cause the sensor to break. This pin is tied up to the 3.3V output of the Arduino.
- GND (Ground)
  Gives a point of reference to the voltages in the circuit. It gets attached to the Arduino GND to complete the circuit.
- SDA (Serial Data Line)
  Transfers information in both directions between the Arduino and the ADXL345. It is linked to the Analog Pin A4 of Arduino, the SDA line of I2C communication.
- SCL (Serial Clock Line)
  Delays the clock signal that is used to synchronize data exchange between the Arduino (master) and ADXL345 (slave). It is tied to Analog Pin A5 on the Arduino.

Arduino Uno R3 Pins

- A4 (SDA)
  Used in the I2C data communication. It sends and receives information to/from the ADXL345.
- A5 (SCL)
  Used for I2C clock generation. The Arduino is the master and determines the communication time.
- 3.3V Pin

The output of the ADXL345 sensor is controlled by supplies to 3.3 V.

- GND Pin
  Provides a common ground between the sensor and the Arduino.
- USB Port (Power Source)
  Power the Arduino directly off the laptop and also allows serial access to sensor data monitoring.

Circuit Operation

- The Arduino Uno will be the I2C master and the ADXL345 will be a slave device. The two communicates via the SDA and SCL lines.
- The Arduino opens the I2C interface and sets the ADXL345 with SparkFun library.
- Arduino transmits instructions to the SDA line which are synchronized by the SCL clock signal.
- The ADXL345 then reacts by sending acceleration data (a x,a y and a z) to the Arduino.
- The Arduino computes this information and serves it to classify the postures.
- Since I2C is only a two-line communication system, the circuit is very simple and small, and this is preferred in the wearable industry.

3.5 Workflow Explanation and System Operation

The PostureSense system operates as a continuous embedded process that acquires sensor data, processes it, and classifies posture in real time. The workflow consists of the following steps:

Step 1: System Initialization

When the Arduino is powered through the USB connection, the system begins execution. The microcontroller initializes:

- Serial communication (for monitoring output)
- I2C communication interface
- ADXL345 sensor using the SparkFun library

The accelerometer is configured for measurement mode and appropriate range settings.

Step 2: Sensor Data Acquisition

The Arduino reads raw acceleration values from the ADXL345:

$$a_x,\ a_y,\ a_z$$

These values represent acceleration along the three axes and are transmitted via the I2C interface. The system samples data at a rate of **10 readings per minute**, which is sufficient for capturing slow posture changes during sleep.

Step 3: Signal Filtering

A **moving average filter** is applied to the raw acceleration data to reduce noise caused by:

- Breathing motion
- Minor body adjustments
- Sensor fluctuations

This step ensures more stable and reliable posture detection.

Step 4: Orientation Estimation

The filtered acceleration values are analyzed to determine the orientation of the body relative to gravity. The dominant axis (highest magnitude) indicates the current posture direction.

Step 5: Posture Classification

The system classifies posture using a threshold-based method:

- $a_z > T \rightarrow$ Supine
- $a_z < -T \rightarrow$ Prone
- $a_x > T \rightarrow$ Right lateral
- $a_x < -T \rightarrow$ Left lateral

This approach is computationally efficient and suitable for real-time embedded systems.

Step 6: Output and Data Logging

The classified posture is:

- Displayed on the **serial monitor**
- Optionally stored or transmitted for further analysis

This allows real-time observation and debugging during development.

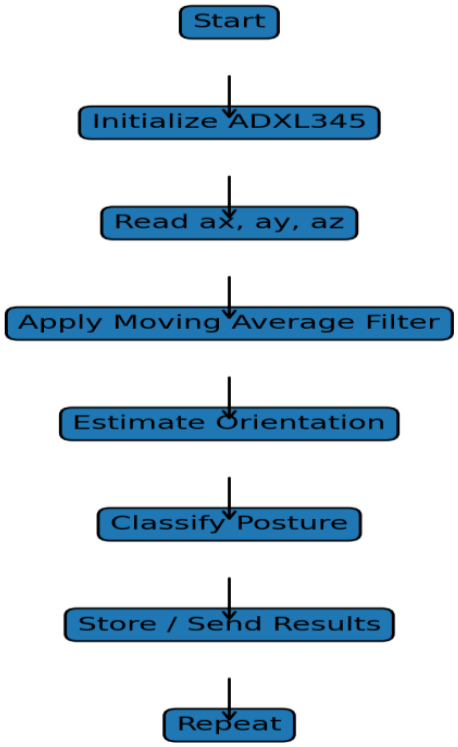
Step 7: Continuous Loop Execution

After classification, the system repeats the process continuously:

Read → Filter → Compute → Classify → Output → Repeat

This loop enables real-time posture tracking throughout the sleep period.

Loop: Initialize → Read → Filter → Classify → Repeat. Sampling = 10 readings/min.



4. Justification of Design Choices

4.1 Accelerometer Range: ±2 g

The ADXL345 offers selectable ranges of ±2 g, ±4 g, ±8 g, and ±16 g. For static tilt detection, the highest sensitivity is desired. At ±2 g, the resolution is 4 mg/LSB (≈ 0.23°), which enables detection of small posture changes [2]. Higher ranges would reduce resolution and increase quantization noise, making precise angle estimation more difficult. Although fast movements could saturate the sensor at ±2 g, such movements are brief and the classification algorithm holds the previous posture during motion, so saturation does not degrade the steady-state classification.

4.2 Communication Protocol: I²C vs. SPI

I²C was selected over SPI for three reasons:

- **Pin count:** I²C uses only two data lines (SDA, SCL), leaving more digital pins available for future expansion (e.g., SD card, Bluetooth). SPI would require CS, MOSI, MISO, and SCLK – four pins.
- **Simplicity:** The SparkFun ADXL345 breakout is already configured for I²C with onboard pull-ups (though external ones are still recommended). The Arduino Wire library is well-supported and requires minimal code.
- **Speed:** The required sampling rate is well within I²C's capability. Even with multiple devices, bus contention is not an issue.

4.3 Sensor Placement

Following the approach of Alinia and Ghasemzadeh [1], the accelerometer is mounted on the chest (mid-sternum). This location provides a consistent relationship between body axes and sensor axes, simplifying classification. The enclosure is attached with medical-grade adhesive patches to ensure stability and comfort during overnight wear.

5. References

[1] P. Alinia and H. Ghasemzadeh, "Pervasive lying posture tracking," *Sensors*, vol. 20, no. 20, 2020.

[2] Analog Devices, "ADXL345 Digital Accelerometer Data Sheet," Rev. E, 2015.

[3] SparkFun Electronics, "ADXL345 Hookup Guide," [Online].
Available: https://learn.sparkfun.com/tutorials/adxl345-hookup-guide.

[4] J. Razjouyan et al., "Improving sleep quality assessment using wearable sensors by including information from postural/sleep position changes and body acceleration," *J. Clin. Sleep Med.*, vol. 13, no. 11, pp. 1301–1310, 2017.